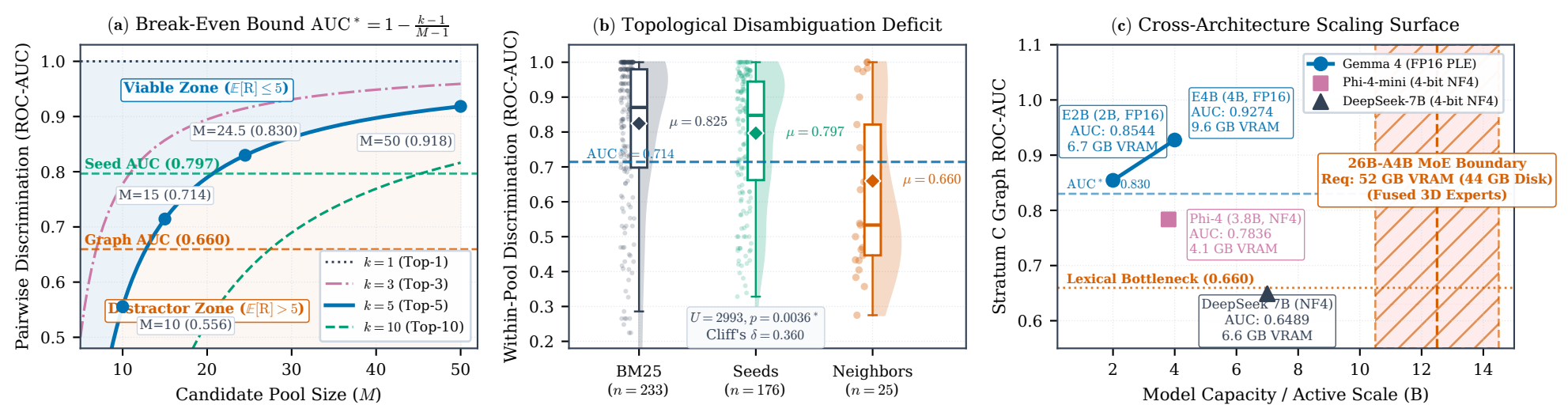

(a) Break-Even Bound AUC* = 1 − (k−1)/(M−1)
Viable Zone (E[R] ≤ 5)
M=24.5 (0.830)
M=50 (0.918)
Seed AUC (0.797)
M=15 (0.714)
Graph AUC (0.660)
Distractor Zone (E[R] > 5)
M=10 (0.556)
k = 1 (Top-1)
k = 3 (Top-3)
k = 5 (Top-5)
k = 10 (Top-10)
Pairwise Discrimination (ROC-AUC)
Candidate Pool Size (M)
(b) Topological Disambiguation Deficit
μ = 0.825
μ = 0.797
μ = 0.660
AUC* = 0.714
U = 2993, p = 0.0036*
Cliff's δ = 0.360
Within-Pool Discrimination (ROC-AUC)
BM25 (n = 233)
Seeds (n = 176)
Neighbors (n = 25)
(c) Cross-Architecture Scaling Surface
Gemma 4 (FP16 PLE)
Phi-4-mini (4-bit NF4)
DeepSeek-7B (4-bit NF4)
E2B (2B, FP16)
AUC: 0.8544
6.7 GB VRAM
E4B (4B, FP16)
AUC: 0.9274
9.6 GB VRAM
AUC* 0.830
26B-A4B MoE Boundary
Req: 52 GB VRAM (44 GB Disk)
(Fused 3D Experts)
Phi-4 (3.8B, NF4)
AUC: 0.7836
4.1 GB VRAM
Lexical Bottleneck (0.660)
DeepSeek-7B (NF4)
AUC: 0.6489
6.6 GB VRAM
Stratum C Graph ROC-AUC
Model Capacity / Active Scale (B)